1.

Let $T(x, y)$ represent the statement "x trusts y" over the domain over the people in the world, and let C be a constant representing "Charlie". Write a statement as a *WFF* in predicate logic.

a. "There is exactly one person who trusts everybody."

$\exists!x.\forall y.T(x, y)$

b. "Charlie trusts at most 2 people."

$\exists!x.\exists!y.(T(C, x) \lor T(C, y) \land x \neq y)$

c. "Everybody is trusted by someone other than themselves."

$\forall x.\exists y.(T(y, x) \land x \neq y)$

d. "There is a person who is trusted by everyone but themselves."

$\exists x.\forall y.(T(y, x) \land y \neq x)$

2.

Translate each formula back into natural language, determining a truth value for each statement.

a. $\forall x.\exists y.(T(x, y) \land \neg T(y, x))$

Everybody trusts someone who doesn't trust them back.

b. $\forall x.\forall y.(T(x, y) \Rightarrow \exists z.(T(x, z) \land T(z, y)))$

If everybody trusts anyone, everybody trusts someone who trusts anyone.

3.

Negate the following and simplify the statements.

a. $\neg\exists x.\forall y.(T(x, y) \Rightarrow T(y, x))$

$\forall x.\exists y.\neg(T(x, y) \Rightarrow T(y, x))$

$\forall x.\exists y.(T(x, y) \land \neg T(y, x))$

b. $\neg\forall x.(T(C, x) \Rightarrow \exists y.(T(x, y))$

$\exists x.\neg(T(C, x) \Rightarrow \exists y.(T(x, y)))$

$\exists x.(T(C, x) \land \neg\exists y.(T(x, y)))$

$\exists x.(T(C, x) \land \forall y.\neg T(x, y))$

4. *Prove the following claim using the direct proof strategy.*

Claim: For any integer n, if n is divisible by 3, then n^2 is divisible by 9.

Let $n = 3m, m \in \mathbb{Z}$

$n^2 = (3m)^2$

$n^2 = 9m^2$

$n^2 = 9(m^2)$

Hence, the claim is $\top$

5. *Prove the following claim in the integer domain using the indirect proof (contrapositive) strategy.*

Claim: If $a + b \geq 100$, then $a \geq 50$ or $b \geq 50$.

Assume that $a + b < 100$ when $a < 50$ and $b < 50$.

$a + b < 50 + 50$

$a + b < 100$

Hence, the claim is $\top$

6. *Prove the following claim using any proof strategy you see fit.*

Claim: For any integer n, if $3n + 7$ is even, then n is odd.

Case 1: n is even:

Let $n = 2m, m \in \mathbb{Z}$

$3n + 7 = 3(2m) + 7$

$= 6m + 7$

$= 2(3m) + 7$

Hence, when n is even, $3n + 7$ is odd

Case 2: n is odd:

Let $n = 2m + 1, m \in \mathbb{Z}$

$3m + 7 = 3(2m + 1) + 7$

$= 6m + 10$

$= 2(3m + 5)$

Hence, when n is odd, $3n + 7$ is odd

Therefore, claim is $\top$

7. *Prove the following claim in the integer domain using proof by contradiction.*

Claim: If $a^2 + b^2$ is odd, then a is even or b is even.

Assume that if $a^2 + b^2$ is odd, a and b are odd.

Let $a, b = 2n + 1, 2m + 1, \{n, m \in \mathbb{Z}\}$

$a^2 + b^2 = (2n + 1)^2 + (2m + 1)^2$

$= 4m^2 + 4n^2 + 4m + 4n + 2$

$= 2(2m^2 + 2n^2 + 2m + 2n + 1)$

Contradiction, hence statement is $\top$

8. *Prove the following claim in the integer domain using case distinction.*

Claim: For each integer n, the remainder of n^2 when divided by 4 is either 0 or 1.

Let the integer r represent the remainder.

Case 1: n is even:

Let $n = 2m, m \in \mathbb{Z}$

$n^2 = 4m^2$

$= 4(m^2)$

$r = 0$

Case 2: n is odd:

Let $n = 2m + 1, m \in \mathbb{Z}$

$n^2 = (2m + 1)^2$

$4m^2 + 4m + 1$

$4(m^2 + m) + 1$

$r = 1$

Therefore, claim is $\top$

9.

For each statement below, decide whether it is true or false. Prove the true statements, choosing your own proof strategy. For the false statements, give a counterexample.

a. “The product of two odd integers is odd.”

Let $x, y = 3, 5$

$x + y = 8$, which is even.

Hence, claim is $\bot$

b. If $x \cdot y$ is even, then x and y are even.

Let $x = 2, y = 5$

$x \cdot y = 10$, yet y is odd.

Hence, claim is $\perp$

c. The sum of two primes is always even.

Let two primes, $x, y = 2, 3$

$x + y = 5$, which is odd.

Hence, claim is $\perp$

d. For all integers n, the term $n^3 + n$ is even.

Case 1: n is even:

Let $n = 2m, m \in \mathbb{Z}$

$n^3 + n = (2m)^3 + 2m$

$= 8m^3 + 2m$

$= 2(4m^3 + m)$

Case 2: n is odd:

Let $n = 2m + 1, m \in \mathbb{Z}$

$n^3 + n = (2m + 1)^3 + (2m + 1)$

$= 8m^3 + 12m^2 + 8m + 2$

$= 2(4m^3 + 6m^2 + 4m + 1)$

Hence, claim is $\top$

5. If n^2 is divisible by 4, then n is divisible by 4.

Let $n = 4m, m \in \mathbb{Z}$

$n^2 = (4m)^2$

$= 16m^2$

$= 4(4m^2)$

Hence, claim is $\top$

10.

Prove each of the following existential statements by providing a witness. State the witness explicitly and verify that it satisfies the predicate.

a. $\exists x : \mathbb{Z}.x^2 = 4x$

Let $x = 4$

$x^2 = 4^2 = 16$

$4x = 4(4) = 16$

$16 = 16\ \{x^2 = 4x\}$

Hence, statement is $\top$

b. $\exists x : \mathbb{Z}, x^3 < x$

Let $x = -2$

$x^3 = (-2)^3 = -8$

$-8 < -2$ $\{x^3 < x\}$

Hence, statement is $\top$

c. $\exists x : \mathbb{Z}.2x + 1 = x - 4$

Let $x = -5$

$2x + 1 = 2(-5) + 1 = -9$

$x - 4 = -5 - 4 = -9$

$-9 = -9$ $\{2x + 1 = x - 4\}$

Hence, statement is $\top$

d. $\exists x : \mathbb{R}.x^2 - 3x + 2 = 0$

Let $x = -2$

$x^2 - 3x + 2 = (-2)^2 - 3(-2) + 2 = 0$

Hence, statement is $\top$

11.

For each statement below, decide whether to disprove it (find a counterexample) or prove it. If disproving, give an explicit counterexample and explain why it works. If proving, use the arbitrary element strategy.

a. $\forall x : \mathbb{Z}.x^2 \geq 2x$

Let $x = 1$

$x^2 = 1^2 = 1$

$2x = 2(1) = 2$

$1 \ngeq 2$ $\{x^2 \ngeq x\}$

Hence, statement is $\bot$

b. $\forall x : \mathbb{R}.x^2 + 1 > 0$

Let x be an arbitrary real number.

$x^2 \geq 0$, as all real numbers squared are non-negative.

Hence, statement is $\top$

c. $\forall x : \mathbb{Z}.x^2 + 2x + 1 \geq 0$

Let x be an arbitrary integer.

$x^2 + 2x + 1 = (x + 1)^2$

$(x + 1)^2 \geq 0$ as all integers squared are non-negative.

Hence, statement is $\top$

d. $\forall x : \mathbb{Z}.x^3 \geq x$

Let $x = -2$

$x^3 = -8$

$-8 \ngeq -2$ $\{x^3 \ngeq x\}$

Hence, statement is $\bot$

12.

For each pair of statements below, decide which are true and which are false over the integers, and justify your answer with a witness, an arbitrary-element argument, or a counterexample.

a. *(1)* $\forall x \in \mathbb{Z}.\exists y \in \mathbb{Z}.x + y = 0$ and

(2) $\exists y \in \mathbb{Z}.\forall x \in \mathbb{Z}.x + y = 0$

(1):

Let x be an arbitrary integer.

A negated form of x, $-x$ exists, such that $x + -x = 0$

Hence, the statement is $\top$

(2):

Let $y = 2$

Only x = −2, satisfies $x + y = 0$

Thus, not all x satisfies the statement

Hence, the statement is $\bot$

b.

(1) $\forall x \in \mathbb{Z}.\exists y \in \mathbb{Z}.y > x$

(2) $\exists y \in \mathbb{Z}.\forall x \in \mathbb{Z}.y > x$

(1):

Let x be an arbitrary integer.

As there is no greatest x, there must exist a y, such that $y > x$

Hence, the statement is $\top$

(2):

Let y be an arbitrary integer.

As there is no greatest integer, there must exist an x, such that $x > y$

Hence, not all x satisfies $y > x$

Thus, the statement is $\bot$

13.

For each of the following, write a formal natural deduction proof. Use only the rules ∀ introduc- tion, ∀ elimination, ⇒ introduction and ⇒ elimination, and state which lines each step depends on.

a. *Given* $\forall x.(C(x) \Rightarrow A(x))$, *and* $C(T)$, *prove* $A(T)$

1	$\forall x.(C(x) \Rightarrow A(x))$	*Premise*
2	$C(T)$	*Premise*
3	$C(T) \Rightarrow A(T)$	$\forall E$ [1]
4	$A(T)$	$\Rightarrow E$ [2, 3]

b. *Given* $\forall x.(P(x) \Rightarrow (Q(x) \Rightarrow R(x)))$ *and* $\forall x.P(x)$, *prove* $\forall x.(Q(x) \Rightarrow R(x))$

1	$\forall x.(P(x) \Rightarrow (Q(x) \Rightarrow R(x)))$	*Premise*
2	$\forall x.P(x)$	*Premise*
3	a	*Assumption*
4	$P(a) \Rightarrow (Q(a) \Rightarrow R(a))$	$\forall E$ [1]
5	$P(a)$	$\forall E$ [2]
6	$Q(a) \Rightarrow R(a)$	$\Rightarrow E$ [4, 5]
7	$\forall x.(Q(x) \Rightarrow R(x))$	$\forall I$ [3 : 6]

c. *Given* $\forall x.(A(x) \Rightarrow B(x))$, *prove* $(\forall x.A(x)) \Rightarrow (\forall x.B(x))$

1	$\forall x.(A(x) \Rightarrow B(x))$	*Premise*
2	$\forall x.A(x)$	*Assumption*
3	a	*Assumption*
4	$A(a)$	$\forall E$ [2]
5	$A(a) \Rightarrow B(a)$	$\forall E$ [1]
6	$B(a)$	$\Rightarrow E$ [4, 5]
7	$\forall x.B(x)$	$\forall I$ [3 : 6]
8	$(\forall x.A(x)) \Rightarrow (\forall x.B(x))$	$\forall I$ [2 : 7]

d. Let $L(x)$: "x is a logician", $M(x)$: "x is methodical" and $H(x)$: "x is happy". Given $\forall x.(L(x) \Rightarrow (M(x) \Rightarrow H(x)))$ and $\forall x.(L(x) \Rightarrow M(x))$, prove $\forall x.(L(x) \Rightarrow H(x))$.

1	$\forall x.(L(x) \Rightarrow (M(x) \Rightarrow H(x)))$	*Premise*
2	$\forall x.(L(x) \Rightarrow M(x))$	*Premise*
3	$\quad a$	*Assumption*
4	$\quad L(a) \Rightarrow M(a)$	$\forall E$ $[2]$
5	$\quad L(a) \Rightarrow (M(a) \Rightarrow H(a))$	$\forall E[1]$
6	$\quad\quad L(a)$	*Assumption*
7	$\quad\quad M(a)$	$\Rightarrow E$ $[4, 6]$
8	$\quad\quad H(a)$	$\Rightarrow E$ $[5, 6]$
9	$\quad L(a) \Rightarrow H(a)$	$\Rightarrow I$ $[6 : 8]$
10	$\forall x.(L(x) \Rightarrow H(x))$	$\forall I$ $[3 : 9]$